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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 15 Turning a light bulb on and off with applause

Sensors and Instrumentation

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1. Introduction

In this practical exercise, a system capable of turning on and off is designed and implemented. A 110V spotlight is activated by applause detection. This is achieved using a chain signal processing that ranges from sound capture to switching
Using a triac with an Arduino Uno, the circuit is expected to:

- Detect the sound of applause and convert it into a processable electrical signal.
- Filter and amplify only the frequency band where applause predominates (3 kHz - 6 kHz).
- Generate an adjustable digital pulse for the Arduino, based on the intensity of the applause.
- Count the number of claps in a 3-second interval to decide when to turn on. (2 claps) or off (4 claps) of the spotlight.
- Switch a triac (2N6071) with an MOC3022 optocoupler to isolate the signal from 110 VAC line control.

The gain is adjusted with a potentiometer in the feedback network (Rf), allowing to control the amplitude of the signal coming from the KY-038 microphone. The circuit with +5V from the Arduino and -5V generated with an L7905 regulator connected to a 9V battery.
The filter was implemented with a second TL081 designed to attenuate all frequencies outside the range of interest, ensuring that only applause generates a significant signal.
A third TL081 functions as a simple window comparator. The trigger threshold is Adjust with another potentiometer so that the output saturates and delivers a digital level (HIGH) when the filtered signal exceeds the reference level.
The comparator output is connected to pin 2 of the Arduino, which detects rising edges. to increase an applause counter.
A 3-second interval is measured (starting with the first clap) within which: 2 claps the Arduino turns on the simulation LED (and sends the signal to the MOC3022 to turn on the triac).

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Four claps turn the Arduino off (and stop triggering the triac). Values exceeding
Applause or the end of the 3 seconds restarts the count. The Arduino will control an MOC3022.
which opto-isolates the control signal and triggers the 2N6071 triac, responsible for switching the light
110 VAC.

2. Definitions and conceptual framework

2.1. KY-038 Microphone

It is a sound sensor module with analog and digital output. In this project, only the sound sensor module is used.
It uses the analog output, which provides a signal proportional to the intensity of the
sound captured. The sensor consists of an electret condenser microphone and a small
integrated amplifier. The signal is weak and requires additional amplification to be useful.
in signal processing.



Figure 1: HC-SR04 Ultrasonic Sensor

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2.2. TL081 operational amplifier

It is a low-noise operational amplifier with differential input and single output.

He used this component for all analog stages of the circuit:

- Initial amplification of the microphone signal
- Bandpass filter
- Level comparator

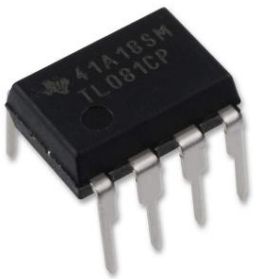


Figure 2: Op Amp TL081

2.3. Active bandpass filter

Applause has a strong energy component between 3kHz and 6kHz. To be sure

that only these frequencies pass through and the system does not react to ambient noise or

For other voices, an active bandpass filter with op amps was designed.

This filter attenuates signals outside that band, dissipating low noise (speech, ambient).

and high (sparking, interference).

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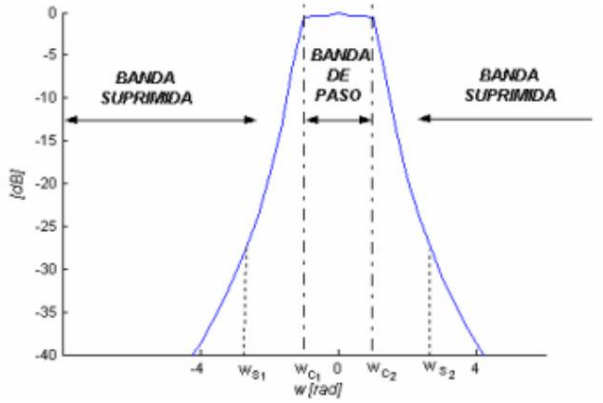


Figure 3: Output of a bandpass filter

2.4. Comparator with adjustable reference level

The filtered signal passes to a comparator built with another TL081, which is compared against an adjustable reference voltage with a potentiometer, when the signal exceeds that When the threshold is reached (by applause), the op amp saturates and its output abruptly changes to a high level (5V), acting as an applause detector.

The output is a digital signal that the Arduino can easily read.

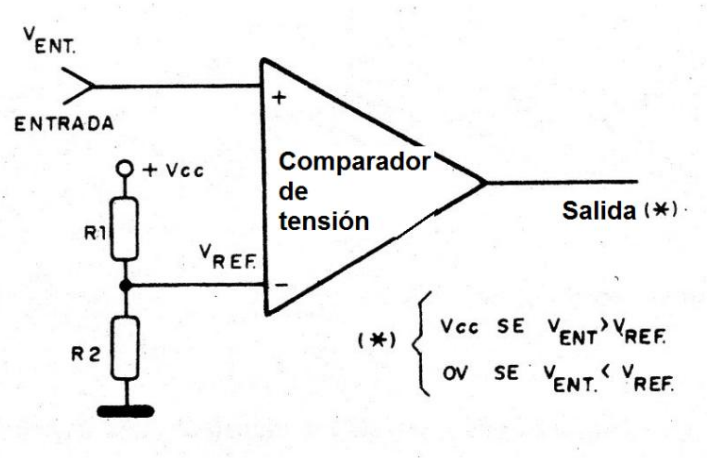


Figure 4: Example of a voltage comparator

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2.5. MOC3022 Optocoupler

The MOC3022 is an optocoupler with TRIAC output designed to control loads
alternating current without the need for a direct connection between the control part and the part
of power. It consists of an internal LED diode which, when activated, triggers a TRIAC
integrated.

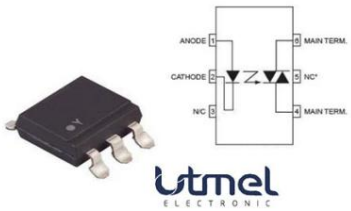


Figure 5: MOC3022 Optocoupler

2.6. TRIAC 2N6071

The 2N6071 is a TRIAC, a semiconductor device that allows current control
It alternates between both polarities of the AC waveform. It functions as a controlled switch.
by a small firing signal on its door

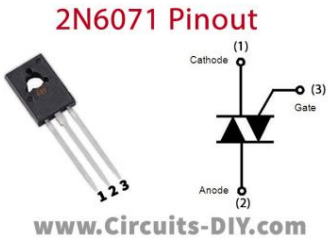


Figure 6: Triac 2N6071

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3. Development

For the circuit design, everything was divided into functional blocks, each with a specific task.
Clear. The first is the sound capture stage, which was done with a KY-038 microphone.
This module has two outputs, one digital and one analog; for this project the following was used:
It's analog because it delivers a signal proportional to the sound intensity. That signal
It is of very low amplitude, barely in millivolts, so a stage was needed
amplification to be able to process it properly.

3.1. Amplification with op amp (TL081)

Amplification was achieved using a TL081 in a non-inverting configuration, powered by
a symmetrical ±5V power supply. We take the +5V directly from the Arduino, and for the
At –5V we use an L7905 regulator with an external –9V source. In this stage also
A precision potentiometer was placed as a variable feedback resistor, for
adjust the amplifier's gain to control how loud a
applause to make it detected.

3.2. Active bandpass filter (3kHz – 6kHz)

Before choosing the frequencies for the filter, it was decided to use the Audacity program.
to analyze the frequencies present in applause, so an applause was recorded and
applied the frequency analysis tool.

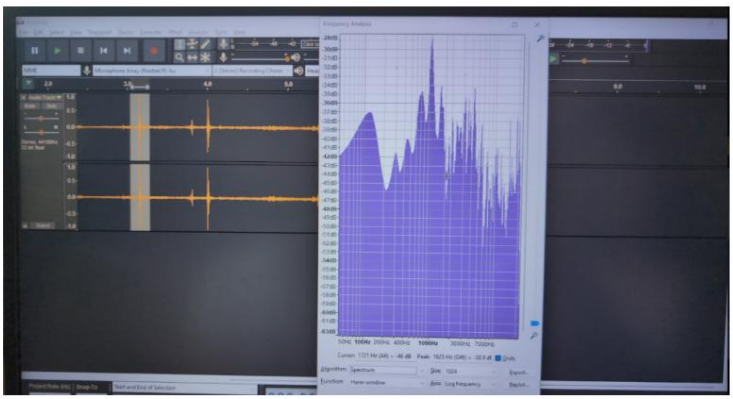


Figure 7: Frequency analysis in Audacity

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Seeing that a large part of the applause frequencies were applied in the same program, a filter with a curve simulating the one that the circuit filter will perform.

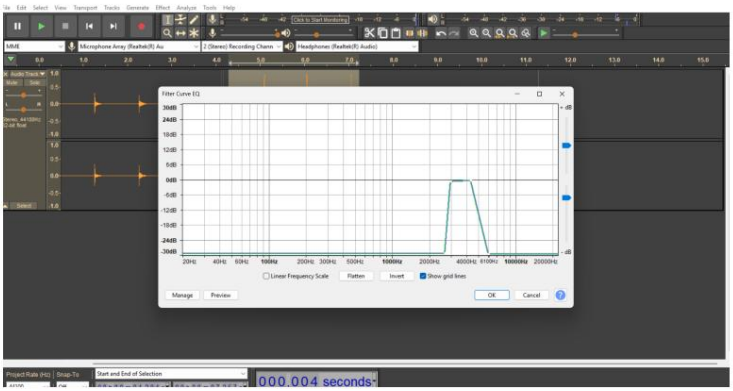


Figure 8: Filter simulation in Audacity

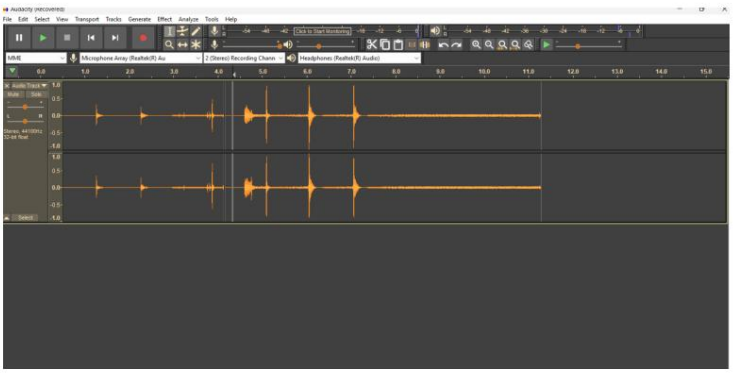


Figure 9: Filter output in Audacity

Once the filter is applied, it can be observed that the other background sounds are filtered out. of the audio and there are some peaks in the audio spectrogram, shown in figure 9 demonstrating that applause prevails after the filter.

Once frequencies for the filter were selected, the design of the filter began.

The capacitor was selected practically, choosing a value of 1nF since it is a an easy-to-obtain value. From that value, the necessary resistances were calculated for center the filter on the mid-frequency of:

$$fr = fH \cdot fL = \ddot{y}\ 6\ kHz \cdot 3\ kHz = 4.242\ kHz$$

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The interim band was defined as:

$$B = fH \ddot{y} fL = 6\ kHz \ddot{y} 3\ kHz = 3\ kHz$$

And the filter quality, such as:

$$Q = \frac{fr}{B} = \frac{4,242}{} = 1,414\ 3$$

With this data, and using the configuration of an active bandpass filter with amplifier
Operationally, the corresponding formulas were used to calculate the values of the
resistors, with the filter constant given by:

$$\frac{1}{RC} = 2\ddot{y}B$$

$$R = \frac{1}{2\ddot{y}BC} \ddot{y} \frac{0.1591}{1nF} \ddot{y}\ 53.033k\ddot{y}\ 3kHz \times$$

The following expression was used for the resistance

$$fr = 2 \frac{\sqrt{1 + \frac{R}{Rr}}}{\ddot{y}\ 2 \times \sqrt{\ddot{y}} \times R \times C}$$

$$Rr = R \left(\frac{fr \cdot RC}{0.1125} \right)^{2} \ddot{y}\ 1^{-1}$$

$$Rr \ddot{y}\ 17,684\ k\ddot{y}$$

3.3. Threshold comparator

After the filter, the signal was sent to a third stage: a voltage comparator made
also with a TL081. Here the filter signal was compared against a reference voltage
adjustable with a potentiometer. If the signal exceeded that threshold, the comparator saturated.
its high voltage (close to 5V), generating a digital output that represented the detection
from a round of applause. This output was connected directly to a digital input of the Arduino.

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One, who was in charge of processing the events. These 3 previous stages were simulated.

in Proteus 10 to observe its operation shown in figure 11.

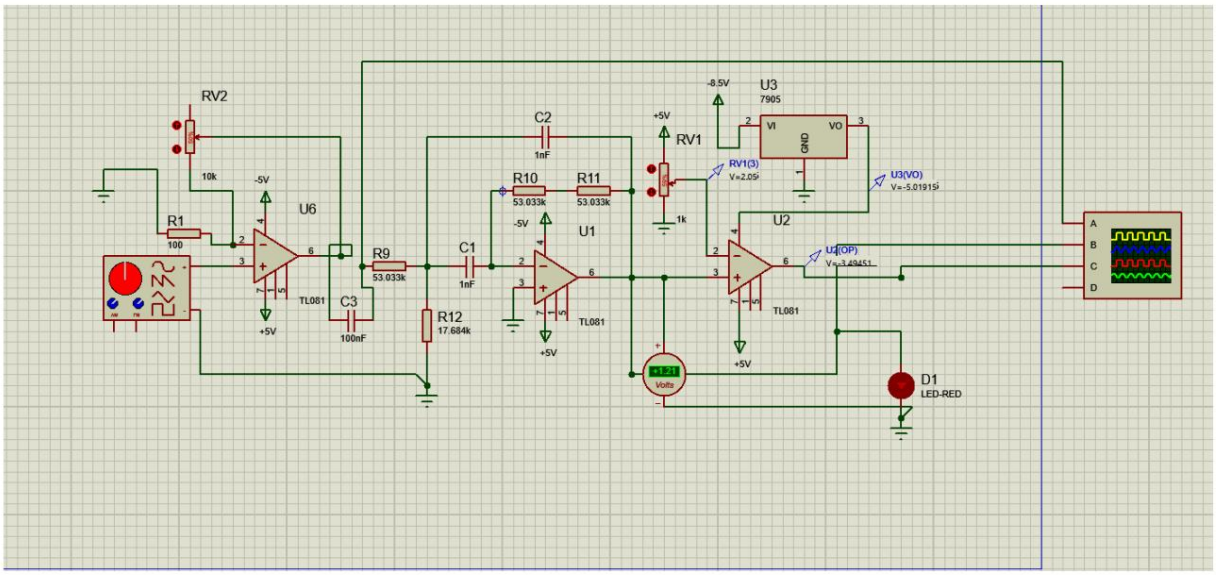


Figure 10: Frequency analysis in Audacity

3.4. Digital Processing

The Arduino was programmed to detect rising edges, that is, when the signal

It went from low to high. Upon detecting applause, it initiated a 3-second window to

count the following.

If there were 2 claps within that time, the light would turn on. If there were 4, it would turn off.

In the event that more than 3 seconds pass without reaching a valid amount or if a valid amount is detected

After more than 4 claps, the counter would automatically reset.

3.5. Power Stage

The final stage was triggering the 110V light bulb with a 2N6071 TRIAC. To isolate the

To control the mains voltage from the Arduino, an MOC3022 was used as an optocoupler.

It sent the signal to the internal LED of the MOC3022, and when it was activated, the TRIAC

It fired, allowing alternating current to pass to the light bulb.

All of this was integrated onto a single punched card with the three TL081s, the MOC3022, the

TRIAC and passive components, leaving the Arduino and the power supply connected by wires.

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4. Code

This was the code used, as explained above:

Listing 1: PID control code for the cart

```
const int pinEntrada = 2;                    // Comparator input
const int pinLED = 4;                        // LED (simulates a light bulb)

int applausecounter = 0;
unsigned long startTime = 0;
bool counting = false ;
bool lightOn = true ;

void setup() {
    pinMode ( pinEntrance ,      INPUT ) ;
    pinMode ( pinLED ,     OUTPUT );
    digitalWrite ( pinLED ,              HIGH);        // The light bulb starts on
    Serial . begin (115200) ;
    Serial. println ("Applause counter started... ");
}

void loop () {
    static bool previousState = LOW ;
    bool currentState = digitalRead(inputPin);

    // Detect rising edge
    if ( currentState == HIGH && previousState == LOW ) {
        if (! counting ) {
            startTime = millis();
            counting = true ;
        }

        applause counter ++;
        Serial.print ("Applause detected! Total: ");
        Serial. println (ApplauseCounter);

        // Immediate actions
```

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```
        if ( applausecounter == 2 && ! lightOn ) {
            digitalWrite ( pinLED ,          HIGH);
            lightOn = true ;
            Serial . println (" >> LIGHT ON <<") ;
        }
        else if ( applausecounter == 4 && lightOn ) {
            digitalWrite ( pinLED ,          LOW ) ;
            lightOn = false ;
            Serial. println (" >> FOCUS OFF <<") ;
        }
        else if (applausecounter > 4) {
            Serial. println (" >> More than 4 claps . Resetting counter . << " ) ;
            applause counter = 0;
            counting = false ;
        }

        delay (200) ; // Anti-bounce
    }

    previousState = currentState;

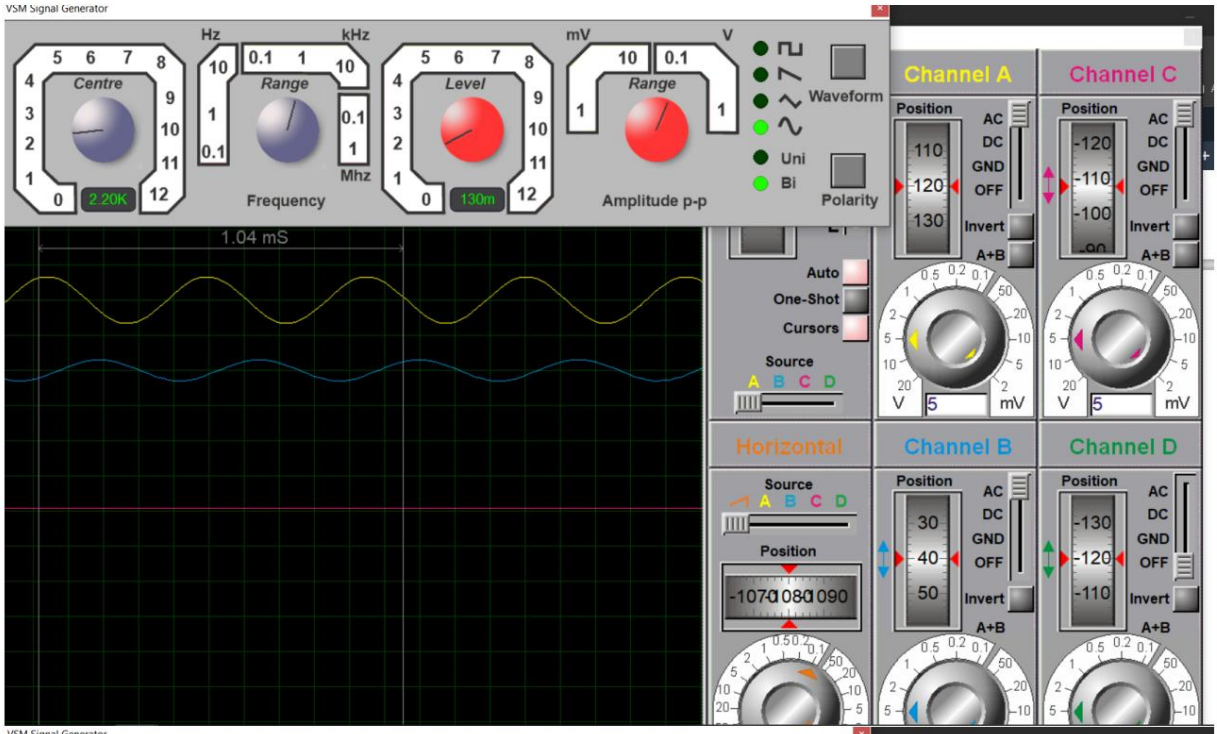
    // If the waiting time expires
    if ( counting && ( millis () - startTime >= 3000) ) {
        Serial . print (" >> Time expired . Counted applause : ") ;
        Serial. print ( applauseCounter ) ;
        Serial. println (". Resetting counter . <<") ;
        applause counter = 0;
        counting = false ;
    }
}
```

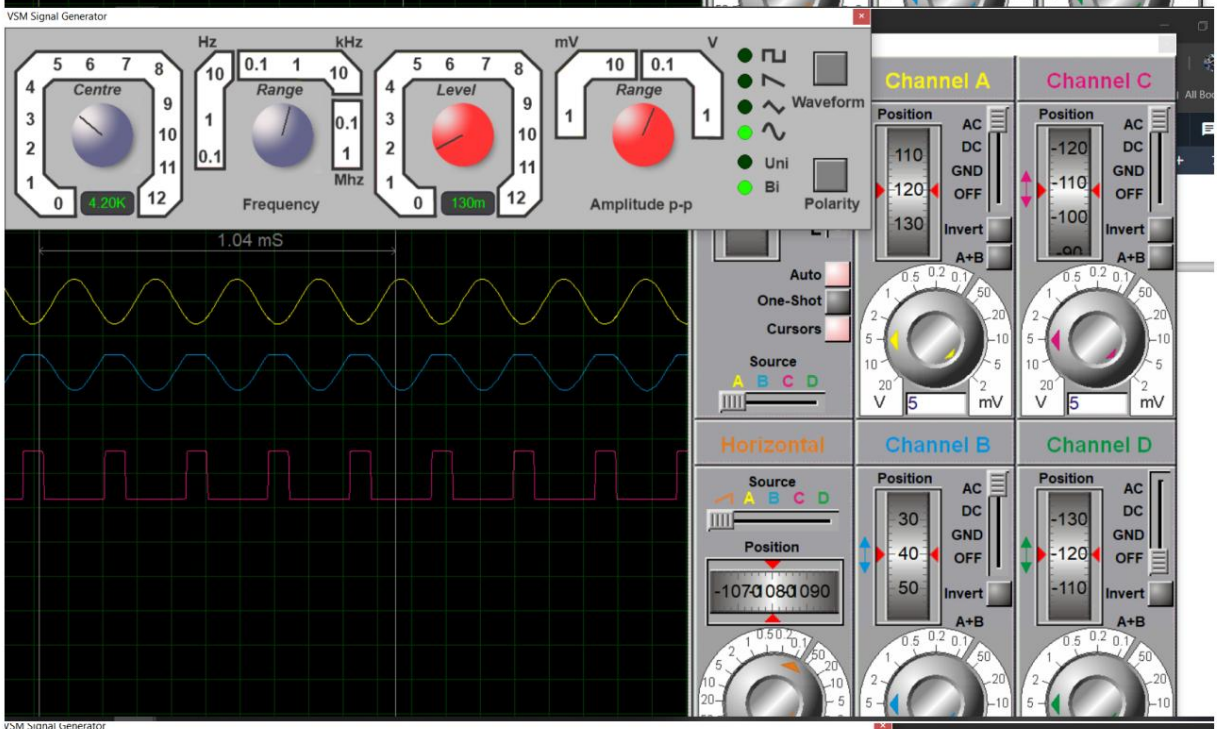
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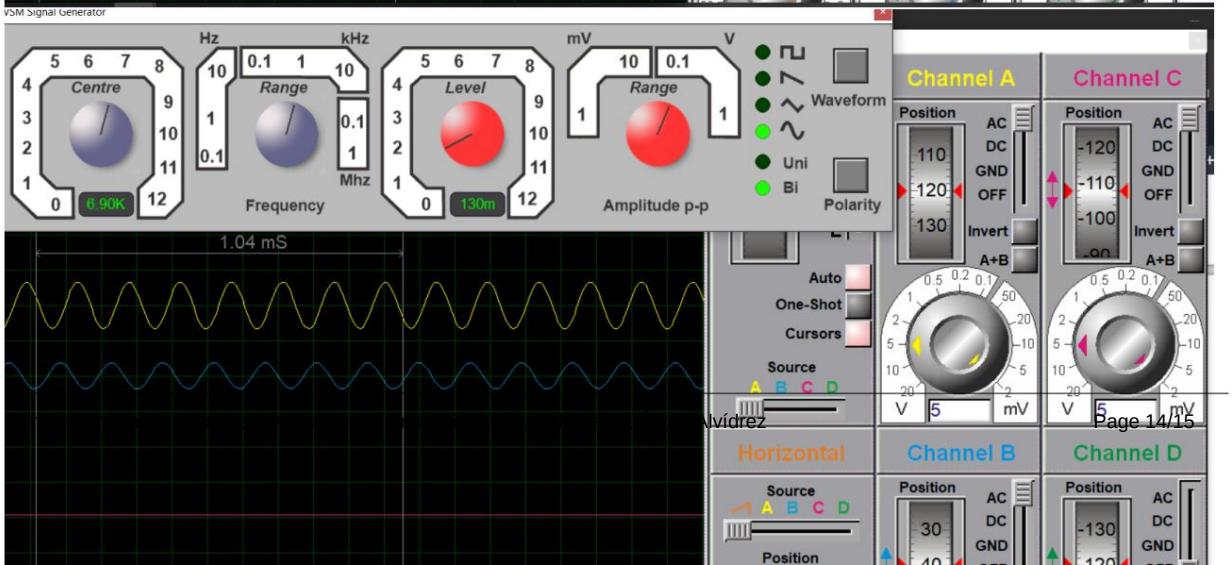
5. Results

The circuit was initially simulated in Proteus, replacing the microphone with a signal generator. In this simulation the behavior was correct: the filter allowed the signal path in the desired range and attenuated out-of-band frequencies as it I expected. However, several details arose during the physical implementation that forced to modify the design. The KY-037 module used as a microphone introduced an offset of approximately 2.5V in the signal, which directly affected the operation of the filter. To mitigate this effect, a ceramic capacitor was incorporated at the input of the circuit in order to block that DC component, an adjustment that was also applied subsequently in the Proteus simulation to reflect the real behavior.

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Furthermore, it was detected that the comparator, in its low output state,
It introduced a small negative voltage that was undesirable at the time of excitation.
the TRIAC gate. This problem was solved by placing a diode that blocks that value
negative and ensures that the control signal remains clean and uninterrupted in the negative state
at rest. With these adjustments, both simulated and practical performance will be improved.
It was stable and the circuit fulfilled its objective of detecting acoustic signals within the
specific frequency range.

6. Conclusion

This project allowed the integration of multiple concepts of analog and digital electronics to
Develop a functional system for detecting specific frequencies using a filter
active bandpass filter with TL081 operational amplifiers and a trigger circuit with
Arduino-controlled TRIAC. Throughout the development process, its importance was verified.
to consider not only the theoretical design, but also the practical details that arise
when implementing real components, such as the unexpected offset of the KY-037 module or the
comparator behavior in low state.
Simulation in Proteus was a key tool for validating the design before the
physical weaponry, but it was also evident that certain elements do not behave in a
identical between the simulation and the practice. This forced direct adjustments to the
circuit, such as including a capacitor to eliminate DC offset or a diode to
Ensure signal levels compatible with TRIAC triggering. These simple solutions
But effective ones show how a good understanding of the behavior of each block is important.
Functional allows you to solve problems without redoing the entire design.
Finally, a system was achieved that detects signals in the desired range and responds
properly switching on an alternating current load, demonstrating practical use
of active filters, comparators and power devices in a control application
automated.